

TITLE

INTRODUCTION OF PROBLEM SOLVING

CSE 1021

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1.INTRODUCTION

The **Personal Expense Tracker** is a Python-based application developed to help users record, organize, and monitor their daily expenses. Managing expenses manually can be difficult and may lead to incorrect calculations or loss of records. This project provides a simple digital solution for maintaining expense information.

The application allows users to enter expense details such as amount, category, description, and date. Users can view their saved expenses, calculate total spending, check category-wise expenditure, search expenses by category, and delete unwanted records.

The project uses **JSON file storage** to save expense information. This allows the data to remain available even after the application is closed and restarted.

2. PROBLEM STATEMENT

Managing daily expenses using notebooks or manual records can become difficult when the number of transactions increases. Users may find it challenging to calculate total spending and identify how much money is spent on different categories.

The purpose of this project is to develop a simple Python application that can digitally record and organize personal expenses while providing basic spending calculations and search functionality.

3. OBJECTIVES

The main objectives of the Personal Expense Tracker are:

1. To create a simple application for recording daily expenses.
 2. To store expense details in an organized format.
 3. To categorize expenses for easier analysis.
 4. To calculate total expenditure automatically.
 5. To display category-wise spending.
 6. To provide an option to search expense records.
 7. To allow users to delete unwanted records.
 8. To store expense information permanently using JSON.
 9. To demonstrate practical Python programming concepts.
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4. SCOPE OF THE PROJECT

The Personal Expense Tracker can be used by students, individuals, and anyone who wants to maintain basic personal expense records.

The current application supports:

- Recording daily expenses.
- Viewing stored expenses.
- Calculating total spending.
- Checking category-wise spending.
- Searching expenses.
- Deleting expense records.
- Storing data locally using JSON.

The project can be further extended with graphical interfaces, databases, reports, charts, and budgeting features.

5. TECHNOLOGIES USED

5.1 Python

Python is used as the main programming language because it provides simple syntax and built-in features suitable for developing this application.

5.2 JSON

JSON is used to store expense records in a structured format. It provides simple local data storage without requiring a separate database.

5.3 File Handling

Python file-handling operations are used to read expense information from the JSON file and save updated records.

5.4 Date and Time

The Python datetime module is used to automatically record the date on which an expense is added.

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop
- Minimum 2 GB RAM
- Basic storage space
- Keyboard and mouse

Software

Requirements

- Windows, Linux, or macOS
- Python 3.x
- Command Prompt or Terminal
- Any Python-compatible editor such as VS Code, PyCharm, or IDLE

No external Python libraries are required.

7. FUNCTIONAL REQUIREMENTS

The system provides the following functions:

Add Expense

The user can enter an expense amount, category, and description. A unique ID and current date are automatically assigned.

View Expenses

The system displays all recorded expenses in a structured format.

Expense Summary

The system calculates the total amount spent and displays the number of recorded expenses.

Category-wise Spending

The application calculates spending for each category.

Search Expense

Users can search for expenses by entering a category.

Delete Expense

Users can delete an existing expense using its unique ID.

Data Storage

All records are stored in the expenses.json file.

8. NON-FUNCTIONAL REQUIREMENTS

The application is designed to be:

- **Simple:** Easy for beginners to understand and use.
- **Reliable:** Stores records in a persistent JSON file.
- **Efficient:** Performs basic calculations quickly.
- **Maintainable:** Uses separate functions for different operations.
- **Portable:** Can run on systems with Python 3 installed.

9. SYSTEM WORKFLOW

The general workflow of the application is:

Start



Load Existing Expense Data



Display Main Menu



Select Operation



Add / View / Search / Summary / Delete



Process User Request



Update JSON Data



Display Result



Return to Main Menu



Exit

10. PROJECT MODULES

The project is mainly implemented in the main.py file.

load_expenses()

Loads previously saved expense records from the JSON file.

save_expenses()

Saves the current expense records into the JSON file.

add_expense()

Accepts expense details from the user and adds a new record. **view_expenses()**

Displays all available expense records. **show_summary()**

Calculates total expenditure and category-wise spending.

search_category()

Searches expense records based on the entered category. **delete_expense()**

Deletes an expense record using its ID.

menu()

Displays the available application options. **main()**

Controls the overall execution of the application.

11. DATA STORAGE

The project uses a file named **expenses.json** for storing data.

A sample record is:

```
{  
  "id": 1,  
  "amount": 250.0,  
  "category": "Food",  
  "description": "Lunch",  
  "date": "2026-09-22"  
}
```

The JSON file allows the application to preserve expense information even after the program is closed.

12. SAMPLE CATEGORIES

Users can maintain expenses under categories such as:

- Food
- Travel
- Shopping
- Bills
- Education
- Entertainment
- Medical

- Other

The user can also enter a custom category when adding an expense.

13. SAMPLE OUTPUT

When the application starts, the following menu is displayed:

=====

PERSONAL EXPENSE TRACKER

=====

1. Add Expense
2. View All Expenses
3. Expense Summary
4. Search by Category
5. Delete Expense
6. Exit

Enter your choice:

Example expense records:

ID	Amount	Category	Description	Date
1	250.00	Food	Lunch	2026-09-22
2	500.00	Travel	Bus Ticket	2026-09-22
3	1200.00	Shopping	Clothing	2026-09-22

14. ADVANTAGES

The main advantages of the project are:

- Easy to use.
- Simple menu-driven interface.
- Helps maintain daily expense records.

- Reduces manual calculations.
- Provides total spending information.
- Provides category-wise spending information.
- Allows searching and deleting records.
- Stores information permanently.
- Does not require an external database.
- Uses standard Python modules.

15. LIMITATIONS

The current version has some limitations:

- It is a command-line application.
- It does not have a graphical user interface.
- It uses a local JSON file instead of a database.
- It does not provide graphical charts.
- It does not include user authentication.
- It does not currently provide monthly or yearly reports.

16. FUTURE ENHANCEMENTS

The project can be enhanced in the future by adding:

1. Graphical User Interface using Tkinter.
 2. Monthly and yearly expense reports.
 3. Budget management.
 4. Income and savings tracking.
 5. Expense charts and graphs.
 6. Excel and PDF report generation.
 7. SQLite or MySQL database integration.
 8. User login and authentication.
 9. Date-based filtering.
 10. Budget limit notifications.
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17. TESTING

The application can be tested using different input scenarios.

Test Case	Input	Expected Result
Add Expense	Valid amount and category	Expense added successfully
Invalid Amount	Text instead of number	Error message displayed
Test Case	Input	Expected Result
View Expenses	Select option 2	All records displayed
Summary	Select option 3	Total and category summary displayed
Search	Enter existing category	Matching expenses displayed
Delete	Enter valid expense ID	Expense deleted
Invalid ID	Enter unavailable ID	Expense not found message
Exit	Select option 6	Application closes

18. LEARNING OUTCOMES

This project helped demonstrate the practical use of several Python programming concepts:

- Variables and data types
- Functions
- Conditional statements
- Loops
- Lists
- Dictionaries
- User input
- Exception handling
- File handling
- JSON data storage
- Date and time operations
- Data processing
- Basic application design

The project also demonstrates how programming concepts can be combined to solve a practical real-world problem.

19. CONCLUSION

The **Personal Expense Tracker** is a simple Python application designed to help users maintain and organize their daily expenses. The system provides essential operations such as adding, viewing, searching, summarizing, and deleting expense records.

The application uses JSON-based file storage to maintain expense information between different program executions. It demonstrates important Python concepts such as functions, lists, dictionaries, file handling, JSON processing, exception handling, and calculations.

The project provides a practical example of how Python can be used to develop a useful application for everyday expense management.

20. REFERENCES

1. Python Official Documentation <https://docs.python.org/>
2. Python JSON Documentation <https://docs.python.org/3/library/json.html>
3. Python datetime Documentation <https://docs.python.org/3/library/datetime.html>